

Jenkins & CI/CD

CI/CD

Continuous Integration (CI) means developers frequently merge code changes into a shared repository, where automated builds and tests validate the changes. Continuous Delivery/Deployment (CD) extends automation so validated software can be released reliably.

Jenkins Architecture

Jenkins has a controller that manages configuration, schedules jobs, stores build history, and presents the web interface. Build work may run on the controller or on separate agents, which let teams scale builds across multiple systems.

Freestyle Jobs

A Freestyle project is a GUI-configured Jenkins job. It can pull source code from GitHub, run shell commands, execute tests, and publish results. It is straightforward for simple automation but is less portable than pipeline code stored in Git.

Declarative Pipelines

A Declarative Pipeline defines CI/CD work in a structured format. It uses stages such as Checkout, Build, and Test, making the build process visible and repeatable.

Jenkinsfile

A Jenkinsfile is a text file stored in the project repository that defines a Jenkins Pipeline. Versioning it with application code allows review, history tracking, and consistent execution across environments.

GitHub Webhooks

A GitHub webhook sends an HTTP request to Jenkins when an event, such as a push, occurs. Jenkins receives the notification through `/github-webhook/` and starts the configured job automatically.

Jenkins Credentials

Jenkins Credentials securely stores secrets such as GitHub tokens, usernames and passwords, SSH keys, and API keys. Credentials should never be committed into a GitHub repository or written directly in a Jenkinsfile; use credential IDs and Jenkins-managed secret storage instead.

Pipeline Stages

- **Checkout:** Retrieves the current source code from GitHub.
- **Build:** Runs commands that prepare or build the application.
- **Test:** Runs automated checks and fails the pipeline if the checks fail.

Jenkins Dashboard

Jenkins

+ New Item

Build History

Build Queue

No builds in the queue.

Build Executor Status

0/2

All

+

S	W	Name ↓	Last Success	Last Failure	Last Duration
✓	☀	freestyle-github-build	1 hr 5 min #1	N/A	5.2 sec
✓	☁	jenkins-ci-pipeline	37 min #2	38 min #1	12 sec

Freestyle project

Jenkins

freestyle-github-build

Workspace of freestyle-git...

Status

Changes

Workspace

Wipe Out Current Workspace

Build Now

Delete Project

GitHub Hook Log

Configure

Rename

Builds

Filter

Today

#1 11:20 AM

freestyle-github-build /

git

app.sh

README.md

test.sh

(all files in zip)

Console Output

Jenkins

freestyle-github-build

#1

Status

Changes

Console Output

Delete build #1

Git Build Data

Edit build information

Download

Copy

View as plain text

Started by user Nisha Subramanian

Running as SYSTEM

Building in workspace /var/lib/jenkins/workspace/freestyle-github-build

The recommended git tool is: NONE

No credentials specified

Cloning the remote Git repository

Cloning repository https://github.com/nisha-subramanian/jenkins-ci-demo.git

> git init /var/lib/jenkins/workspace/freestyle-github-build # timeout=10

Fetching upstream changes from https://github.com/nisha-subramanian/jenkins-ci-demo.git

> git --version # timeout=10

> git --version # 'git version 2.43.0'

> git fetch --tags --force --progress -- https://github.com/nisha-subramanian/jenkins-ci-demo.git +refs/heads/*:refs/remotes/origin/* # timeout=10

> git config remote.origin.url https://github.com/nisha-subramanian/jenkins-ci-demo.git # timeout=10

> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10

Avoid second fetch

> git rev-parse refs/remotes/origin/main^{commit} # timeout=10

Checking out Revision 317414b5b1f6d4d6f346a32977b5e9ab337362 (refs/remotes/origin/main)

> git config core.sparsecheckout # timeout=10

> git checkout -f 317414b5b1f6d4d6f346a32977b5e9ab337362 # timeout=10

Commit message: "Add Jenkins CI demo"

First time build. Skipping changelog.

[freestyle-github-build] \$ /bin/sh -xe /tmp/jenkins17256338685733479552.sh

+ chmod +x app.sh test.sh

+ ./app.sh

Application build completed successfully.

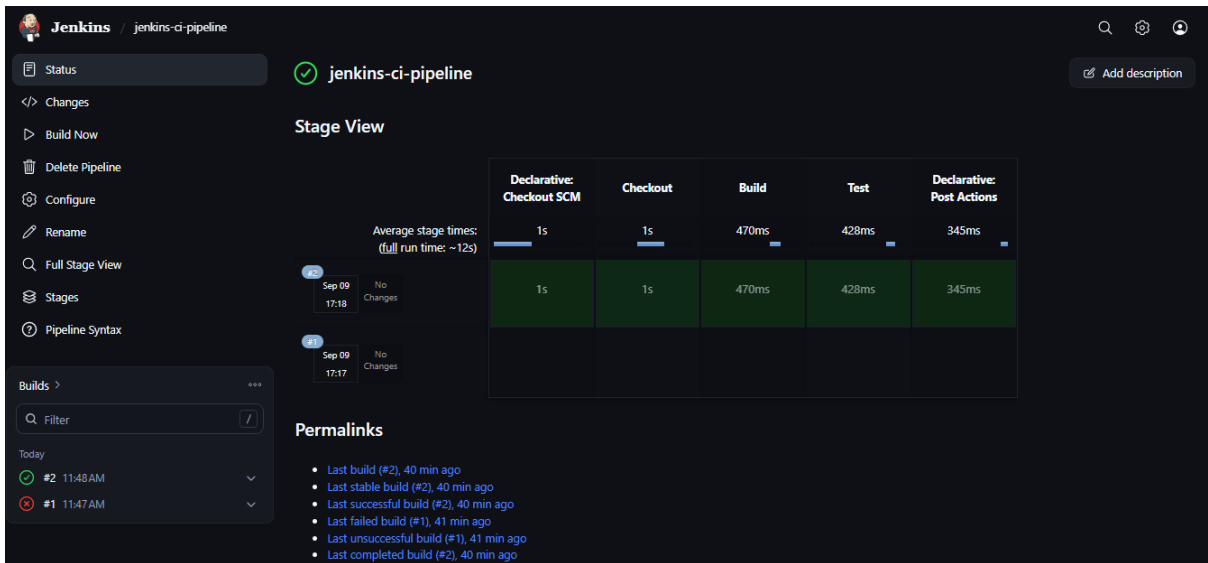
+ ./test.sh

Running tests...

All tests passed.

Finished: SUCCESS

Pipeline Execution



The Jenkins interface for the 'jenkins-ci-pipeline' shows a successful execution status. The left sidebar contains navigation options: Status, Changes, Build Now, Delete Pipeline, Configure, Rename, Full Stage View, Stages, and Pipeline Syntax. The main area displays the 'Stage View' with a table of stage execution times and a 'Permalinks' section.

jenkins-ci-pipeline

Stage View

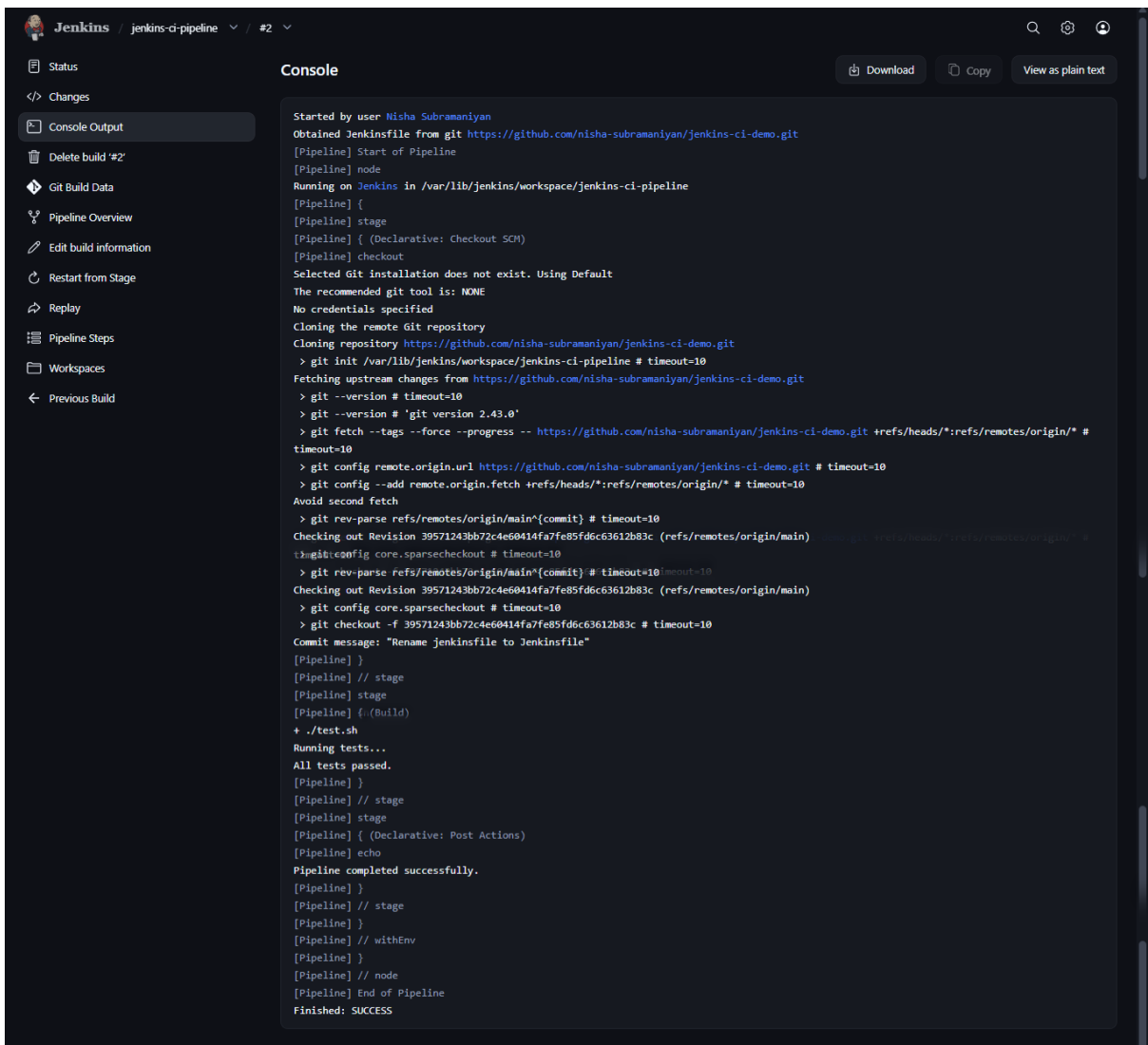
Average stage times: (full run times: ~12s)

Stage	Declarative: Checkout SCM	Checkout	Build	Test	Declarative: Post Actions
1	1s	1s	470ms	428ms	345ms
2	1s	1s	470ms	428ms	345ms

Permalinks

- Last build (#2), 40 min ago
- Last stable build (#2), 40 min ago
- Last successful build (#2), 40 min ago
- Last failed build (#1), 41 min ago
- Last unsuccessful build (#1), 41 min ago
- Last completed build (#2), 40 min ago

Console Output



The Jenkins interface for the 'jenkins-ci-pipeline' shows the console output for build #2. The left sidebar contains navigation options: Status, Changes, Console Output, Delete build #2, Git Build Data, Pipeline Overview, Edit build information, Restart from Stage, Replay, Pipeline Steps, Workspaces, and Previous Build. The main area displays the console output text.

jenkins-ci-pipeline #2

Console

Started by user Nisha Subramanian

Obtained Jenkinsfile from git https://github.com/nisha-subramanian/jenkins-ci-demo.git

[Pipeline] Start of Pipeline

[Pipeline] node

Running on Jenkins in /var/lib/jenkins/workspace/jenkins-ci-pipeline

[Pipeline] {

[Pipeline] stage

[Pipeline] { (Declarative: Checkout SCM)

[Pipeline] checkout

Selected Git installation does not exist. Using Default

The recommended git tool is: NONE

No credentials specified

Cloning the remote Git repository

Cloning repository https://github.com/nisha-subramanian/jenkins-ci-demo.git

> git init /var/lib/jenkins/workspace/jenkins-ci-pipeline # timeout=10

Fetching upstream changes from https://github.com/nisha-subramanian/jenkins-ci-demo.git

> git --version # timeout=10

> git --version # 'git version 2.43.0'

> git fetch --tags --force --progress -- https://github.com/nisha-subramanian/jenkins-ci-demo.git +refs/heads/*:refs/remotes/origin/* # timeout=10

> git config remote.origin.url https://github.com/nisha-subramanian/jenkins-ci-demo.git # timeout=10

> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10

Avoid second fetch

> git rev-parse refs/remotes/origin/main^{commit} # timeout=10

Checking out Revision 39571243bb72c4e60414fa7fe85fd6c63612b83c (refs/remotes/origin/main)

> git config core.sparsecheckout # timeout=10

> git rev-parse refs/remotes/origin/main^{commit} # timeout=10

Checking out Revision 39571243bb72c4e60414fa7fe85fd6c63612b83c (refs/remotes/origin/main)

> git config core.sparsecheckout # timeout=10

> git checkout -f 39571243bb72c4e60414fa7fe85fd6c63612b83c # timeout=10

Commit message: "Rename jenkinsfile to Jenkinsfile"

[Pipeline] }

[Pipeline] // stage

[Pipeline] stage

[Pipeline] { (Build)

+ ./test.sh

Running tests...

All tests passed.

[Pipeline] }

[Pipeline] // stage

[Pipeline] stage

[Pipeline] { (Declarative: Post Actions)

[Pipeline] echo

Pipeline completed successfully.

[Pipeline] }

[Pipeline] // stage

[Pipeline] }

[Pipeline] // withEnv

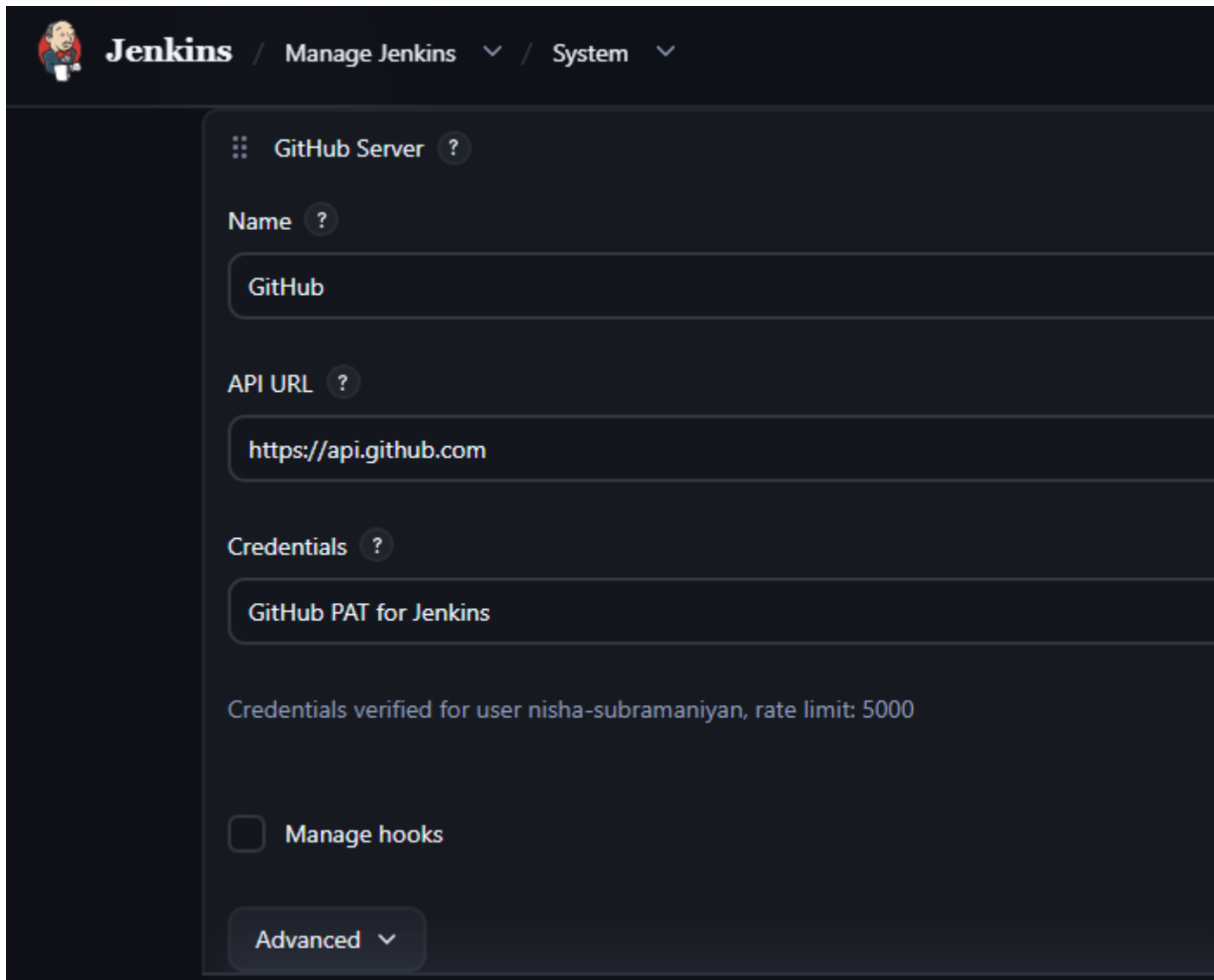
[Pipeline] }

[Pipeline] // node

[Pipeline] End of Pipeline

Finished: SUCCESS

GitHub Integration



The image shows the Jenkins web interface for configuring a GitHub server. The top navigation bar includes the Jenkins logo and links to 'Manage Jenkins' and 'System'. The main content area is titled 'GitHub Server' with a help icon. It contains several input fields: 'Name' (filled with 'GitHub'), 'API URL' (filled with 'https://api.github.com'), and 'Credentials' (filled with 'GitHub PAT for Jenkins'). Below these fields, a status message reads 'Credentials verified for user nisha-subramaniyan, rate limit: 5000'. At the bottom, there is an unchecked checkbox for 'Manage hooks' and an 'Advanced' dropdown menu.

Jenkins / Manage Jenkins / System

GitHub Server ?

Name ?

GitHub

API URL ?

https://api.github.com

Credentials ?

GitHub PAT for Jenkins

Credentials verified for user nisha-subramaniyan, rate limit: 5000

☐ Manage hooks

Advanced ▾

GitHub Link: <https://github.com/nisha-subramaniyan/jenkins-ci-demo.git>